Discover the Power of Java Heap and Thread Dump Tools

In the earlier article published in the December 2013 issue of *OSFY*, the author covered the basics like dump types, dump formats, modes of acquiring dumps, the terminology associated with dumps, etc. With this grounding the reader is well equipped to delve further into the topic.

In this article, we explore the usage options and features available with tools from Oracle. JDK is shipped with various command line tools and a visual tool jvisualvm that is shipped with JDK 1.6, update 7. In this part, we examine jvisualvm since it is most commonly used and provides much better user interface than command line tools. Command line tools need to be used in situations where remote connection to production machines is not possible and those tools are covered in a subsequent part.

Oracle JVisualVM

JVisualVM is a memory and thread analysis tool that is shipped along with the JDK itself. This tool can be considered a combination of all command line tools. It does not work with IBM JVMs but only with Oracle JVMs. This tool is shipped by default from Java version 1.6 update 7 and is available in the JDK bin directory. JVisualVM also can be downloaded from *http://visualvm.java.net*. Various plug-ins are available for use with this tool. One of the most useful plug-ins is the Visual Garbage Collection (VisualGC) plug-in that can be used to visually observe the different generations on the heap and monitor their sizes, in real time. JVisualVM can access and display information about processes running locally and Java processes running on remote hosts.

Troubleshooting start-up issues

If JVisualVM does not start, you can try the following options.

1) Run it from the command line and set an explicit JDK:

```
C:\visualvm_135\bin\visualvm --jdkhome c:\jdk1.7.0_07
```

2) After starting, if there is an error message that states: 'Local Java Applications cannot be monitored', the reason could be a case mismatch between the login user name and the user name in the default application directory. To solve this, go to the system temp directory such as *C:\Users\<user-name>\AppData\Local\Temp\ or C:\Temp* and delete the directory *hsperfdata*

Overview

Once JVisualVM is open, it automatically monitors the local machine and lists any newly started Java processes. Information about the Java process can be obtained by opening that process.